

Terbutaline: Drug information - UpToDate

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For additional information see "Terbutaline: Patient drug information" and "Terbutaline: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Prolonged tocolysis:

Terbutaline has not been approved and should not be used for prolonged tocolysis (beyond 48 to 72 hours). In particular, terbutaline should not be used for maintenance tocolysis in the outpatient or home setting. Serious adverse reactions, including death, have been reported after administration of terbutaline to pregnant women. In the mother, these adverse reactions include increased heart rate, transient hyperglycemia, hypokalemia, cardiac arrhythmias, pulmonary edema, and myocardial ischemia. Increased fetal heart rate and neonatal hypoglycemia may occur as a result of maternal administration.

Brand Names: Canada

- Bricanyl Turbuhaler

Pharmacologic Category

- Antidote, Extravasation;
- Beta₂ Agonist

Dosing: Adult

Asthma, refractory or severe exacerbation

Asthma, refractory or severe exacerbation (alternative agent):

Note: Rarely used and based on limited data; may use in patients whose symptoms are refractory to standard therapy and who are in impending respiratory arrest. May also consider in patients with severe exacerbations who are unable to use inhaled bronchodilators (Ref). Do not use in combination with parenteral epinephrine (Ref).

SUBQ: Initial: 0.25 mg once; if severe symptoms persist after initial dose, may repeat every 20 minutes for a total of 3 doses (Ref).

Asthma, intermittent symptom relief

Asthma, intermittent symptom relief (alternative agent):

Note: An alternative short-acting inhaled beta agonist or rapid-onset long-acting beta agonist/inhaled steroid combination is preferred. Use on an as-needed basis (reliever therapy) rather than regularly scheduled (Ref).

Dry powder inhaler (0.5 mg/actuation) (Canadian product): **Oral inhalation:** One inhalation every 4 hours as needed; if dose is not effective after 5 minutes, may repeat. If second dose is not effective, consult health care provider immediately (maximum dose: 6 inhalations/day). **Note:** If adequate relief is not obtained with previously effective dose or duration of effect is <3 hours, patient should be reassessed promptly; may indicate worsening asthma (Ref).

Oral: 5 mg 3 times daily (~6-hour intervals); if side effects occur, reduce dose to 2.5 mg 3 times daily; maximum daily dose: 15 mg/day. **Note:** Clinical practice guidelines do not recommend oral short-acting beta-agonists or nonselective beta-agonists (including oral terbutaline) for routine management and treatment of asthma due to slower onset of action and higher risk of adverse effects (Ref).

Extravasation management, sympathomimetic vasoconstrictors

Extravasation management, sympathomimetic vasoconstrictors (off-label use):

Large extravasations: **SUBQ:** Infiltrate affected extravasation area with terbutaline 1 mg using a solution of terbutaline 1 mg diluted in 10 mL NS (administration volume varied from 3 to 10 mL); may repeat dose after 15 minutes (Ref).

Small/distal extravasations: **SUBQ:** Infiltrate affected extravasation area with terbutaline 0.5 mg using a solution of terbutaline 1 mg diluted in 1 mL NS; may repeat dose after 15 minutes (Ref).

Tocolysis

Tocolysis (off-label use):

Note: Only for acute, short-term use (≤ 72 hours).

IV: 2.5 to 5 **mcg**/minute; increase gradually every 20 to 30 minutes by 2.5 to 5 **mcg**/minute up to a maximum of 25 **mcg**/minute; decrease to the lowest effective dose once contractions are controlled (Ref).

SUBQ: 0.25 mg every 20 minutes to 3 hours; hold for pulse >120 beats per minute. Terbutaline has not been approved for and should not be used for prolonged tocolysis (beyond 48 to 72 hours) (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, studies suggest no need for adjustment in kidney impairment (Ref)

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Terbutaline: Pediatric drug information")

Asthma

Asthma:

Acute exacerbation:

Emergency room management: Limited data available:

Note: Continuous cardiac monitoring is recommended (Ref).

Children: SUBQ: 0.01 mg/kg/dose every 20 minutes for 3 doses; may repeat every 2 to 6 hours as needed; maximum dose: 0.4 mg/dose; some experts recommend administering every 10 to 15 minutes (Ref).

Adolescents: SUBQ: 0.25 mg/dose; may repeat every 20 minutes for up to 3 doses (Ref); some experts recommend a higher dose (0.4 mg) every 10 to 15 minutes (Ref).

Intensive care unit management: Limited data available; optimal dose not defined; efficacy results variable.

Note: Continuous cardiac monitoring is recommended (Ref).

Children and Adolescents: Continuous infusion: Initial: IV bolus dose: 4 to 10 mcg/kg followed by continuous infusion of 0.1 to 0.4 mcg/kg/minute, titrate by 0.1 to 0.2 mcg/kg/minute increments as frequently as every 30 minutes based on patient response or toxicity. Usual maximum dose is 2 to 3 mcg/kg/minute; however, doses as high as 10 mcg/kg/minute have been described; monitor closely for adverse reactions (Ref).

Acute symptom relief:

Oral: **Note:** Although an FDA-approved indication, guidelines do not recommend oral terbutaline for routine treatment of asthma (Ref).

Children ≥ 12 and Adolescents < 15 years: Oral: 2.5 mg three times daily; maximum daily dose: 7.5 mg per **24 hours** (Ref).

Adolescents ≥ 15 years: Oral: 5 mg three times daily (~ every 6 hours); reduce dose to 2.5 mg three times daily if side effects occur; maximum daily dose: 15 mg per **24 hours** (Ref).

Oral inhalation: *Canadian labeling:* Bricanyl Turbuhaler [Canadian product]: Children ≥ 6 years and Adolescents: 1 inhalation (0.5 mg) as needed; if not effective after 5 minutes may repeat dose. If second dose is not effective, consult health care provider immediately. Maximum daily dose: 6 inhalations per **24 hours**.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%:

Central nervous system: Nervousness, restlessness

Endocrine & metabolic: Decreased serum potassium, increased serum glucose

Neuromuscular & skeletal: Tremor

1% to 10%:

Cardiovascular: Hypertension, tachycardia

Central nervous system: Dizziness, drowsiness, headache, insomnia

Dermatologic: Diaphoresis

Gastrointestinal: Dysgeusia, nausea, vomiting, xerostomia

Neuromuscular & skeletal: Muscle cramps, weakness

<1%, postmarketing, and/or case reports: Cardiac arrhythmia, chest pain, hyperglycemia (preterm labor), hypokalemia (preterm labor), hypotension (preterm labor), ischemic heart disease (preterm labor), lactic acidosis (Smith 2019), myocardial infarction (preterm labor), paradoxical bronchospasm, pulmonary edema (preterm labor)

Contraindications

Hypersensitivity to terbutaline, sympathomimetic amines, or any component of the formulation.

Injection: Additional contraindications: Prolonged tocolysis (>48 to 72 hours), especially for maintenance in the outpatient setting.

Oral: Additional contraindications: Acute or maintenance tocolysis.

Canadian labeling: Additional contraindications (not in the US labeling): Hypersensitivity to lactose; history of tachyarrhythmias; as tocolytic in patients at risk of premature labor or threatened abortion.

Warnings/Precautions

Concerns related to adverse effects:

- **Bronchospasm:** Paradoxical bronchospasm that may be life-threatening may rarely occur with use of inhaled bronchodilating agents; this reaction should be distinguished from inadequate response. Discontinue immediately if paradoxical bronchospasm occurs and institute alternative therapy.
- **Hypersensitivity reactions:** Immediate hypersensitivity reactions (urticaria, angioedema, rash, bronchospasm) have been reported.
- **Serious effects/fatalities:** Do not exceed recommended dose or frequency; serious adverse events, including fatalities, have been associated with excessive use of inhaled sympathomimetics.

Disease-related concerns:

- **Asthma:** Appropriate use: When used as a bronchodilator, optimize anti-inflammatory treatment before initiating maintenance treatment with terbutaline. Do not use as a component of chronic therapy without an anti-inflammatory agent.
- **Cardiovascular disease:** Use with caution in patients with cardiovascular disease (arrhythmia, coronary insufficiency, hypertension, heart failure); beta-agonists may cause elevation in blood pressure, heart rate and result in CNS stimulation/excitation. Beta₂-agonists have been reported to produce ECG changes, such as flattening of the T wave, prolongation of the QTc interval, and ST segment depression.
- **Diabetes:** Use with caution in patients with diabetes mellitus; beta₂-agonists may increase serum glucose and aggravate preexisting diabetes and ketoacidosis.
- **Glaucoma:** Use with caution in patients with glaucoma; may elevate intraocular pressure.
- **Hyperthyroidism:** Use with caution in hyperthyroidism; may stimulate thyroid activity.
- **Hypokalemia:** Use with caution in patients with hypokalemia; beta₂-agonists may transiently decrease serum potassium.
- **Seizures:** Use with caution in patients with seizure disorders; beta-agonists may result in CNS stimulation/excitation.

Dosage form specific issues:

- **Lactose;** Bricanyl Turbuhaler [Canadian product]: Dry powder inhalers may contain lactose; hypersensitivity reactions have been reported in patients with milk protein allergy.

Other warnings/precautions:

- **Patient information:** Patients must be instructed to seek medical attention in cases where acute symptoms are not relieved or a previous level of response is diminished. The need to increase frequency of use may indicate deterioration of asthma, and treatment must not be delayed.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as sulfate:

Generic: 1 mg/mL (1 mL)

Tablet, Oral, as sulfate:

Generic: 2.5 mg, 5 mg

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Terbutaline Sulfate Injection)

1 mg/mL (per mL): \$1.56 - \$24.00

Tablets (Terbutaline Sulfate Oral)

2.5 mg (per each): \$5.44 - \$13.73

5 mg (per each): \$6.65 - \$16.77

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Aerosol Powder Breath Activated, Inhalation:

Bricanyl Turbuhaler: 0.5 mg/actuation (1 ea) [contains lactose monohydrate]

Extemporaneous Preparations

A 1 mg/mL oral suspension may be made with tablets. Crush twenty-four 5 mg tablets in a mortar and reduce to a fine powder. Add 5 mL purified water USP and mix to a uniform paste; mix while adding simple syrup, NF in incremental proportions to **almost** 120 mL; transfer to a calibrated bottle, rinse mortar with vehicle, and add quantity of simple syrup, NF sufficient to make 120 mL. Label "shake well" and "refrigerate". Stable for 30 days (Ref).

Administration: Adult

Oral inhalation: Dry powder inhaler: Bricanyl Turbuhaler [Canadian product]: When ready to use, remove lid and hold inhaler upright. Turn blue grip as far as it will go in one direction, then turn blue grip in the opposite direction until you hear a click; repeat this step a second time and then inhaler is ready to use. Exhale fully (not into inhaler) and place lips around mouthpiece and inhale deeply; remove inhaler from mouth prior to exhaling. Rinse mouth with water after use (do not swallow). If inhaler is dropped or shaken, or if patient exhales into the inhaler after a dose is loaded, the dose will be lost and a new dose should be loaded and inhaled. Clean mouthpiece once weekly with a dry tissue; do not use water or any other fluid to clean and keep inhaler dry. First appearance of red mark in dose indicator indicates that 10 doses remain; discard inhaler when the dose indicator reaches "0".

SUBQ: Administer injections into the lateral deltoid area.

Extravasation management, sympathomimetic vasopressors (off-label use): Stop vesicant infusion immediately and disconnect IV line (leave needle/cannula in place); gently aspirate extravasated solution from the IV line (do **NOT** flush the line); remove needle/cannula; elevate extremity (Ref).

Large extravasations: **SUBQ:** Infiltrate affected extravasation area with terbutaline 1 mg using a solution of terbutaline 1 mg diluted in 10 mL NS (administration volume varied from 3 to 10 mL); may repeat dose after 15 minutes (Ref).

Small/distal extravasations: **SUBQ:** Infiltrate affected extravasation area with terbutaline 0.5 mg using a solution of terbutaline 1 mg diluted in 1 mL NS; may repeat dose after 15 minutes (Ref).

Preparation for Administration: Adult

For extravasation management (off-label use): Using vial for injection, dilute 1 mg in 10 mL (large extravasation site) **or** 1 mg in 1 mL (small/distal extravasation site) of 0.9% sodium chloride (Ref).

Administration: Pediatric

Oral: Tablet: May administer without regard to food; administer around-the-clock to promote less variation in peak and trough serum levels.

Parenteral:

SUBQ: Administer undiluted into the lateral deltoid area.

IV:

Direct IV injection: Administer undiluted over 5 to 10 minutes

Continuous IV infusion: Administer via an infusion pump at a concentration ≤ 1 mg/mL (Ref).

Inhalation: Bricanyl Turbuhaler [Canadian product]: To prepare inhaler prior to use, load dose by holding inhaler in upright position and turn blue grip as far as it will go in one direction and then turn it as far as it will go in the other direction. Prior to first use, this procedure should be done twice; with subsequent dosing, perform this procedure once. Clicking sound means inhaler is loaded with dose and ready for use. Place mouthpiece between teeth and close lips over mouthpiece. Inhale deeply and forcefully. Do not exhale through inhaler. If the Turbuhaler is dropped, shaken, or breathed into after it is loaded, the dose will be lost and a new dose will need to be loaded. When a red mark appears in the dose indicator window, 20 doses are left. When the red mark reaches the bottom of the window, the inhaler should be discarded.

Preparation for Administration: Pediatric

Parenteral: Continuous IV infusion: Pediatric patients (asthma): May dilute in D5W or NS to a concentration < 1 mg/mL (Ref)

Storage/Stability

Injection: Store vial at 20°C to 25°C (68°F to 77°F); protect from light by storing in original carton until use.

Dry powder inhaler (Bricanyl Turbuhaler [Canadian availability]): Store at 15°C to 30°C (59°F to 86°F).

Use: Labeled Indications

Asthma/Bronchospasm: Prevention and reversal of bronchospasm in patients ≥ 12 years of age with asthma and reversible bronchospasm associated with bronchitis and emphysema **or** ≥ 6 years of age (Bricanyl Turbuhaler [Canadian product]).

Note: Asthma (oral product): Although included as an FDA-approved use in the manufacturer's prescribing information for asthma and reversible bronchospasm associated with bronchitis and emphysema, oral short-acting beta-agonists or nonselective beta-agonists (including oral terbutaline) are not recommended due to slower onset of action and higher risk of adverse effects (GINA 2024; NAEPP 2007).

Use: Off-Label: Adult

Extravasation management, sympathomimetic vasoconstrictors; Tocolysis

Medication Safety Issues

Sound-alike/look-alike issues:

Brethine may be confused with Methergine

Terbutaline may be confused with terbinafine, TOLBUTamide

Terbutaline and methylergonovine parenteral dosage forms look similar. Due to their contrasting indications, use care when administering these agents.

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Amisulpride (Oral): May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk C: Monitor*

Antidiabetic Agents: Hyperglycemia-Associated Agents may decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Atomoxetine: May increase hypertensive effects of Sympathomimetics. Atomoxetine may increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Atomoxetine: May increase tachycardic effects of Beta2-Agonists. Atomoxetine may increase hypertensive effects of Beta2-Agonists. *Risk C: Monitor*

Atosiban: Beta2-Agonists may increase adverse/toxic effects of Atosiban. Specifically, there may be an increased risk for pulmonary edema and/or dyspnea. *Risk C: Monitor*

Azithromycin (Systemic): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Azithromycin (Systemic). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Beta-Blockers (Beta1 Selective): May decrease bronchodilatory effects of Beta2-Agonists. *Risk C: Monitor*

Beta-Blockers (Nonselective): May decrease bronchodilatory effects of Beta2-Agonists. Management: Patients with asthma or COPD receiving beta-2 agonists should not normally be treated with non-selective beta-blockers. If used concomitantly, monitor closely for diminished bronchodilatory effects of the beta2-agonist. *Risk D: Consider Therapy Modification*

Bitter Orange: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Bornaprine: Sympathomimetics may increase anticholinergic effects of Bornaprine. *Risk C: Monitor*

Cannabinoid-Containing Products: May increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Carbetocin: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Carbetocin. Management: Monitor for QTc interval prolongation and ventricular arrhythmias

when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Chloroquine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Chloroquine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Clofazimine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Clofazimine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Cocaine (Topical): May increase hypertensive effects of Sympathomimetics. Management: Consider alternatives to use of this combination when possible. Monitor closely for substantially increased blood pressure or heart rate and for any evidence of myocardial ischemia with concurrent use. *Risk D: Consider Therapy Modification*

Dabrafenib: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Dabrafenib. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Dihydralazine: Sympathomimetics may decrease therapeutic effects of Dihydralazine. *Risk C: Monitor*

Domperidone: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Domperidone. *Risk X: Avoid*

Doxapram: Sympathomimetics may increase adverse/toxic effects of Doxapram. *Risk C: Monitor*

Doxofylline: Sympathomimetics may increase adverse/toxic effects of Doxofylline. *Risk C: Monitor*

Droxidopa: Sympathomimetics may increase hypertensive effects of Droxidopa. *Risk C: Monitor*

Encorafenib: May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Esketamine (Injection): May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for elevated heart rate, hypertension, and arrhythmias may be increased. *Risk C: Monitor*

Etelcalcetide: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Etelcalcetide. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Fexinidazole: May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Fluorouracil Products: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Fluorouracil Products. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Gadobenate Dimeglumine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Gadobenate Dimeglumine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Guanethidine: May increase hypertensive effects of Sympathomimetics. Guanethidine may increase arrhythmogenic effects of Sympathomimetics. *Risk C: Monitor*

Halofantrine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Halofantrine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Haloperidol: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Haloperidol. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Inotuzumab Ozogamicin: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Inotuzumab Ozogamicin. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Kratom: May increase adverse/toxic effects of Sympathomimetics. *Risk X: Avoid*

Levoketoconazole: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Levoketoconazole. *Risk X: Avoid*

Linezolid: May increase hypertensive effects of Sympathomimetics. Management: Consider initial dose reductions of sympathomimetic agents, and closely monitor for enhanced blood pressure elevations, in patients receiving linezolid. *Risk D: Consider Therapy Modification*

Lofexidine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Lofexidine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Loop Diuretics: Beta2-Agonists may increase hypokalemic effects of Loop Diuretics. *Risk C: Monitor*

Loxapine: Agents to Treat Airway Disease may increase adverse/toxic effects of Loxapine. More specifically, the use of Agents to Treat Airway Disease is likely a marker of patients who are likely at a greater risk for experiencing significant bronchospasm from use of inhaled loxapine. Management: This is specific to the Adasuve brand of loxapine, which is an inhaled formulation. This does not apply to non-inhaled formulations of loxapine. *Risk X: Avoid*

Meglumine Antimoniate: May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Methacholine: Beta2-Agonists (Short-Acting) may decrease therapeutic effects of Methacholine. Management: Hold short-acting beta₂ agonists for 6 hours before methacholine use. *Risk D: Consider Therapy Modification*

Midostaurin: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Midostaurin. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Monoamine Oxidase Inhibitors: May increase adverse/toxic effects of Beta2-Agonists. *Risk C: Monitor*

Ondansetron: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Ondansetron. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Oxytocin: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Oxytocin. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when

these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Pentamidine (Systemic): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Pentamidine (Systemic). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Pimozide: May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk X: Avoid*

Piperaquine: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Piperaquine. *Risk X: Avoid*

Probucol: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Probucol. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Agents (Highest Risk): Terbutaline may increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this combination. Patients with other risk factors (eg, older age, female sex, bradycardia, hypokalemia, hypomagnesemia, heart disease, and higher drug concentrations) are likely at greater risk for these toxicities. *Risk D: Consider Therapy Modification*

QT-prolonging Antidepressants (Moderate Risk): May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Antipsychotics (Moderate Risk): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Antipsychotics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Class IC Antiarrhythmics (Moderate Risk): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Class IC Antiarrhythmics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-Prolonging Inhalational Anesthetics (Moderate Risk): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Kinase Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Miscellaneous Agents (Moderate Risk): Terbutaline may increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk) may increase serum concentrations of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when

these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Quinolone Antibiotics (Moderate Risk): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Quinolone Antibiotics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Strong CYP3A4 Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Sertindole: May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk X: Avoid*

Solriamfetol: Sympathomimetics may increase hypertensive effects of Solriamfetol. Sympathomimetics may increase tachycardic effects of Solriamfetol. *Risk C: Monitor*

Sotalol: May decrease therapeutic effects of Terbutaline. Terbutaline may increase QTc-prolonging effects of Sotalol. *Risk X: Avoid*

Succinylcholine: Terbutaline may increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Sympathomimetics: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Tedizolid: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for increased blood pressure and heart rate may be increased. *Risk C: Monitor*

Theophylline Derivatives: May increase hypokalemic effects of Beta2-Agonists. Beta2-Agonists may increase adverse/toxic effects of Theophylline Derivatives. Specifically, sympathomimetic effects may be increased. *Risk C: Monitor*

Thiazide and Thiazide-Like Diuretics: Beta2-Agonists may increase hypokalemic effects of Thiazide and Thiazide-Like Diuretics. *Risk C: Monitor*

Thioridazine: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Thioridazine. *Risk X: Avoid*

Thyroid Products: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk of hypertension, tachycardia, and coronary insufficiency may be increased in patients with coronary artery disease. *Risk C: Monitor*

Toremifene: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Toremifene. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Tricyclic Antidepressants: May increase adverse/toxic effects of Beta2-Agonists. *Risk C: Monitor*

Pregnancy Considerations

Terbutaline crosses the placenta; umbilical cord concentrations are ~11% to 48% of maternal blood levels.

Terbutaline may affect uterine contractility; use caution if needed to control bronchospasm in pregnant patients.

Uncontrolled asthma is associated with adverse events in pregnancy (increased risk of perinatal mortality, preeclampsia, preterm birth, low-birth-weight infants, cesarean delivery, and the development of gestational diabetes). Poorly controlled asthma or asthma exacerbations may have a greater fetal/maternal risk than what is associated with appropriately used asthma medications. Maternal treatment improves pregnancy outcomes by reducing the risk of some adverse events (eg, preterm

birth, gestational diabetes). Although short acting beta-2 agonists should be used to treat acute asthma exacerbations in pregnant patients, agents other than terbutaline are preferred (ERS/TSANZ [Middleton 2020]; GINA 2024).

Terbutaline injection has not been approved for and should not be used for prolonged tocolysis (beyond 48 to 72 hours). Oral terbutaline sulfate has not been approved and should not be used for acute or maintenance tocolysis. Terbutaline should not be used for maintenance tocolysis in the outpatient or home setting. Serious adverse reactions, including death, have been reported after administration of terbutaline to pregnant patients. Maternal adverse reactions include increased heart rate, transient hyperglycemia, hypokalemia, cardiac arrhythmias, pulmonary edema, and myocardial ischemia. Increased fetal heart rate and neonatal hypoglycemia may also occur. Terbutaline has been used in the management of preterm labor. Tocolytics may be used for the short-term prolongation of pregnancy to allow for the administration of antenatal steroids and should not be used prior to fetal viability or when the risks of use to the fetus or mother are greater than the risk of preterm birth (ACOG 2016).

Breastfeeding Considerations

Terbutaline is present in breast milk.

Terbutaline breast milk concentrations are similar to or higher than those in the maternal plasma. Based on information from four cases, exposure to the breastfed infant would be <1% of the weight-adjusted maternal dose. Adverse events were not observed in nursing infants (Boréus 1982; Lönnholm 1982).

The manufacturer recommends that terbutaline be used in breastfeeding patients only if the potential benefit to the mother outweighs the possible risk to the infant.

Terbutaline is considered probably acceptable for use while breastfeeding (ERS/TSANZ [Middleton 2020]).

Monitoring Parameters

FEV₁, peak flow, and/or other pulmonary function tests; shortness of breath; BP, heart rate; CNS stimulation; serum glucose, serum potassium; monitor for signs and symptoms of pulmonary edema (when used as a tocolytic). If used for extravasation management, monitor and document extravasation site.

Mechanism of Action

Relaxes bronchial and uterine smooth muscle by action on beta₂-receptors with less effect on heart rate

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Oral: 30 to 45 minutes; SubQ: 6 to 15 minute; Inhalation: 5 minutes (maximum effect: 15 to 60 minutes).

Duration: Oral: 4 to 8 hours; Oral inhalation: 3 to 6 hours; SubQ: 1.5 to 4 hours.

Absorption: 33% to 50%.

Protein binding: 25%.

Metabolism: Hepatic to inactive sulfate conjugates.

Bioavailability: SubQ doses are more bioavailable than oral.

Half-life elimination: 5.7 hours (range: 2.9 to 14 hours).

Time to peak, serum: SubQ: 0.5 hours.

Excretion: Urine (~60% as unchanged drug); feces.

Patient Counseling Points

(For additional information see "Terbutaline: Patient drug information")

What is this drug used for?

- It is used to open the airways in lung diseases where spasm may cause breathing problems.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Fatigue
- Restlessness
- Tremors

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- High blood sugar like confusion, fatigue, increased thirst, increased hunger, passing a lot of urine, flushing, fast breathing, or breath that smells like fruit.
- Low potassium like muscle pain or weakness, muscle cramps, or an abnormal heartbeat.
- Chest pain
- Fast heartbeat
- Abnormal heartbeat
- Severe anxiety
- Severe dizziness
- Passing out
- Severe headache
- Vision changes
- Seizures
- Trouble breathing
- Wheezing
- Cough
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Bricanyl;
- (AT) Austria: Bricanyl | Bricanyl-duriles;
- (AU) Australia: Bricanyl;
- (BD) Bangladesh: Tervent;
- (BE) Belgium: Bricanyl;
- (BR) Brazil: Bricanyl | Greencanyl;
- (CH) Switzerland: Bricanyl;
- (CN) China: Bricanyl | Hui bang | Terbutaline | Terbutaline Sulphate;
- (CO) Colombia: Bricanyl | Sulter | Terburop | Terbutalina;
- (CZ) Czech Republic: Bricanyl;
- (DE) Germany: Asthmoprotect | Bricanyl | Contimit | Terbul | Terbutalin al | Terbutalin ratiopharm;
- (DO) Dominican Republic: Terburop;
- (EC) Ecuador: Bricanyl | Terburop;
- (EE) Estonia: Bricanyl;
- (EG) Egypt: Aironyl | Bricanyl | Terbutaline | Terbutanyl;
- (ES) Spain: Tedipulmo | Terbasmin | Terbasmin ern;
- (FI) Finland: Bricanyl;
- (FR) France: Bricanyl | Bricanyl Ip;
- (GB) United Kingdom: Bricanyl | Monovent;
- (GR) Greece: Dracanyl;
- (HK) Hong Kong: Apt Terbutaline | Ataline | Bricanyl | Britaline | Brothines | Butaline | Butylin | Dhatalin | Eurobutal | Terbron | Terbuline | Terbuta | Terbutaline | Tertal | Uni-Breth;
- (HU) Hungary: Bricanyl;
- (ID) Indonesia: Asmabet | Asthmoprotect | Bintasma | Brasmatic | Bricasma | Forasma | Lasmalin | Nairret | Neosma | Prosmalin | Pulmobron | Sedakter | Tabas | Terasma | Tismalin | Yarisma;
- (IE) Ireland: Bricanyl;
- (IL) Israel: Bricalin;
- (IN) India: Bricanyl | Terbutaline Sulphate | Tetrasma;
- (IT) Italy: Terbasmin;
- (JO) Jordan: Bricanyl | Talin;
- (JP) Japan: Bricanyl | Convon | Naropatin;
- (KR) Korea, Republic of: Beta-2 | Bricanyl;
- (KW) Kuwait: Bricanyl | Butaline;
- (LB) Lebanon: Bricanyl;
- (LT) Lithuania: Bricanyl;
- (LU) Luxembourg: Bricanyl | Terbul;
- (LV) Latvia: Bricanyl;
- (MA) Morocco: Bricanyl;
- (MX) Mexico: Bricanyl | Taziken;
- (MY) Malaysia: Anvelin | Asmalin | Ataline | Bricanyl | Britaline | Bronkine | Brontaline | Bucanil | Butaline | Butanil | Butylin | Pritalin | Terbron | Terbuline | Terbutaline | Terbutaline Sulphate;
- (NL) Netherlands: Bricanyl;
- (NO) Norway: Bricanyl | Terbutaline lannett;
- (NZ) New Zealand: Bricanyl | Terbutalin al;
- (PH) Philippines: Astebroton | Blucodil | Bricanyl | Bronchodam | Lungex | Pulmoxcel | Terbulin | Terbutaline | Terbutaline Am-Europharma | Terbutaline Scheele | Ventryl;
- (PK) Pakistan: Asmanyil | Brethin | Bricanyl | Respair | Terbulin | Terbutascot | Terbutil | Tesmic | Ventair;
- (PL) Poland: Bricanyl;
- (PR) Puerto Rico: Brethine | Terbutaline;

- (PT) Portugal: Bricanyl;
- (QA) Qatar: Bricanyl | Dilanyl | Talin | Terbu-Q;
- (RO) Romania: Bricanyl;
- (RU) Russian Federation: Aironyl | Bricanyl;
- (SA) Saudi Arabia: Bricanyl;
- (SE) Sweden: Bricanyl | Respirol;
- (SG) Singapore: Anvelin | Asmalin | Ataline | Bricanyl | Bucanil | Butylin | Dhatalin | Terbutaline;
- (SI) Slovenia: Bricanyl;
- (SK) Slovakia: Bricanyl;
- (TH) Thailand: Asmaline | Asmarin | Asmotab | Asthamsian | Asthermine | Asthmic | Asthmotab | B-lene | Baltic | Becasyl | Biterine | Bricanyl | Britaline | Broncho | Broncholine | Bronchonil | Bronchonyl | Bronco Asmo | Bronic | Bucanyl | Bucaril | Buline | Butalin | Butaline | Butanyl | Byterine | Cabrinyll | Cocolax | Fasma | Framagon | H ma | Manotaline | P-Canil | Piptanyl | Proasma-t | Terbaline | Terbron | Terbu | Terbulin | Terbuline | Terbumed | Terbuno | Terbusin | Terbutab | Terbutal | Terbutaline | Terbuton | Terlain | Tolbin | Unires;
- (TN) Tunisia: Bricanyl | Talin;
- (TR) Turkey: Bricanyl;
- (TW) Taiwan: Berocol | Brandyll | Bricanyl | Bricardyl | Brothine | Glin | Soter | Terbeline | Terbuline | Tersultran | Tusoloc;
- (UA) Ukraine: Bricanyl;
- (VE) Venezuela, Bolivarian Republic of: Bricanyl | Nortol;
- (ZA) South Africa: Bricanyl

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